

Legal Text Classification of U.S. Supreme Court Decisions

Two-page project write-up | Project ID: PES2UG24AM045

PROJECT GOAL

Automatically assign a Supreme Court decision to a broad legal issue area using its written opinion text.

1. Problem statement

U.S. Supreme Court opinions are long, specialized documents. This project explores whether a machine-learning pipeline can use the opinion text to predict the case's high-level issue-area label. The intended output is a category that helps organize or browse a corpus; it is not a legal judgment or a substitute for legal research.

2. Dataset details

The notebook uses textacy's *SupremeCourt* dataset: approximately 8,400 modern-era U.S. Supreme Court decisions from November 1946 through June 2016. Each record provides decision text and metadata, including issue and issue-area codes [1]. In the notebook run, 8,419 records were loaded; records with a missing issue or an unknown issue-area code were removed, leaving 8,396 labeled texts across 14 issue areas.

The labels are imbalanced: the largest recorded issue area contains 1,924 cases, while one area contains only one. This imbalance makes a single accuracy number incomplete and makes performance on rare classes uncertain.

3. Approach and method

Opinion text is lowercased, whitespace is normalized, and punctuation is removed. The notebook compares three supervised pipelines: (1) TF-IDF features capped at 5,000 terms, reduced to 14 latent topics with Latent Dirichlet Allocation (LDA), then classified with Logistic Regression; (2) the same topic features with class-balanced Logistic Regression; and (3) 50-dimensional Doc2Vec document embeddings followed by Logistic Regression. A fixed 80/20 train/test split with random state 42 is used in the notebook.

How the pipeline works

01 TEXT	02 REPRESENT	03 CLASSIFY	04 EVALUATE
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Implementation, results & conclusion

What was built, what the notebook reported, and what to improve next.

4. Brief implementation

The notebook downloads records through textacy, filters unlabeled examples, cleans the text, constructs model inputs, trains the three classifiers, and prints classification reports. Its final demo cleans a sample legal passage, obtains a Doc2Vec vector, and predicts an issue-area name. The prepared project script and README provide local setup instructions and compute metrics from predictions rather than hard-coding summary values.

5. Results recorded in the notebook

Model	Accuracy	Weighted F1
TF-IDF + LDA + Logistic Regression	44.82%	0.36
Class-weighted TF-IDF + LDA + Logistic Regression	23.87%	0.22
Doc2Vec + Logistic Regression	66.49%	0.66

Interpretation: Doc2Vec + Logistic Regression had the strongest reported result in this notebook run. Class weighting reduced overall accuracy and weighted F1 in the recorded comparison. These are the notebook's recorded metrics, not results from a fresh run of the cleaned project script.

Evaluation caveat

The notebook trains Doc2Vec on all 8,396 documents before creating its train/test split. The test documents therefore influence the learned document representation, which weakens the independence of the holdout set and may make the 66.49% score optimistic. The notebook also reports several F1 summaries as fixed constants. The prepared script instead splits before fitting representations and computes its metrics directly; its results should be run and reported separately.

6. Conclusion

The project demonstrates an end-to-end legal-text classification workflow and suggests that document embeddings may capture useful signals for assigning broad issue areas. The recorded Doc2Vec result is promising as an initial experiment, but it is not a clean generalization estimate because of representation leakage. Before presenting the score as final, rerun with a stratified split made before all feature learning, tune and evaluate with cross-validation, and report per-class precision, recall, and F1 alongside accuracy. The very small classes need particular care.

Dataset source

[1] textacy 0.11.0, "Supreme Court decisions" dataset documentation. https://textacy.readthedocs.io/en/0.11.0/api_reference/datasets.html
Dataset description and date range are from the documentation; record counts and model scores above are transcribed from the supplied Colab notebook PDF.